

Kandinsky As You Preferred

Large Text-to-Image Model Personalization with Semantic Injection

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ABSTRACT

Due to the significant generative capabilities of GenAI (generative artificial intelligence), the art community has actively embraced it to create painterly content. Large text-to-image models can quickly generate aesthetically pleasing outcomes. However, the process can be non-deterministic and often involves tedious trial-and-error as users struggle to formulate effective prompts to achieve their desired results. This paper describes a generative approach that empowers users to easily work with a large text-to-image (TTI) model to create their preferred painterly content. The authors propose a large model personalization method, namely *Semantic Injection*, to personalize a large TTI model in a given specific artistic style, i.e., Kandinsky's paintings in Bauhaus era, as the *Artist Model*. Through working with a Kandinsky expert, the authors first establish a semantic descriptive guideline and a TTI dataset of Kandinsky style and then apply the *Semantic Injection* method to obtain an *Artist Model* of Kandinsky, empowering users to create preferred Kandinsky content in a deterministically controllable manner.

CCS CONCEPTS

• Computing methodologies → Artificial intelligence.

KEYWORDS

GenAI, Large Model Personalization, Semantic Injection

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1 INTRODUCTION

GenAI has been developed for various media contents, including text, sounds, images, and videos, demonstrating the potential to facilitate human creativity. Recently, the text-to-image art (TTI-art)

community has been actively engaging with GenAI, such as Stable Diffusion and Midjourney. Given the growing capabilities of GenAI and the friendly nature of the TTI-art community, digital art created with large models is becoming a mainstream phenomenon.

Prompt engineering, prompt programming, or prompting, allows users to generate content without in-depth knowledge of underlying technologies. However, interactions between users and AI are not always centered around human experience. Due to text inputs' unstructured and open-ended nature, the process often becomes a brute-force trial and error. In other words, the process can be non-deterministic; prompting remains an ongoing challenge for users.

Chaos in the context of digital art, particularly in generative art, is not always viewed as a negative phenomenon but rather as opportunities that artists prefer to work with [McCormack et al. 2014]. Benjamin Kovach describes generative art as a continual search for the ideal balance between complete control and total chaos. To substantiate the control of non-determinism in GenAI from an artistic standpoint, we explore how to utilize large TTI models for painterly content generation in a controllable manner and propose to use *Semantic Injection* to personalize an *artist model*.



Figure 1: Prompting with our *Artist Model* in deterministically controllable manner

We summarize this paper's contributions as (1) a large text-to-image model personalization method-*Semantic Injection* and a personalized *Artist Model* empowering users to create preferred Kandinsky content in a deterministically controllable manner (as in Fig.1) and (2) a text-to-image dataset of Wassily Kandinsky's Bauhaus style; both are open-sourced¹.

2 SEMANTIC INJECTION

Fig.2 shows our proposed method-*Semantic Injection* on personalizing a large text-to-image model for artistic content generation.

¹Kandinsky as preferred: <https://github.com/aaaven/kandinsky-as-preferred/>

Initially, we establish a semantic descriptive guideline of a given artistic style (i.e., Kandinsky Bauhaus style). Built upon [Hu et al. 2021]’s research of Low-Rank Adaptation of Large Language Models (LoRA), the TTI-art community has dedicated substantial efforts and explored several effective variations. We delve into these community efforts and adapt effective ones like Fast LoRA² and DiffLoRA³ to inject previously not recognized features to the large text-to-image model as *Semantic Injection*.

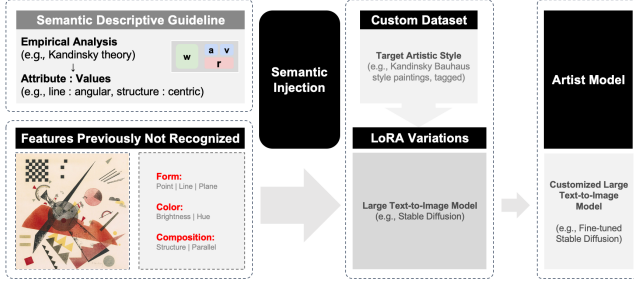


Figure 2: How *Semantic Injection* personalizes large TTI model

Fast LoRA technique adjusts transformer model’s attention layers by adding low-rank matrices to the model’s weights to achieve significant changes in model behavior by modifying a small number of parameters. For the pre-trained weight matrix $W \in \mathbb{R}^{n \times m}$, the adjusted weight matrix can be represented as $W' = W + \Delta W = W + AB^T$, where $A \in \mathbb{R}^{n \times d}$ and $B \in \mathbb{R}^{m \times d}$, n is the dimension of the original weight matrix and d is the low-rank factor and smaller than d . Notably, residual ΔW is decomposed into the product of low-rank matrices A and B with much smaller size, allowing efficient model tuning instead of updating larger W . Furthermore, this technique only fine-tune parts of the transformer model (i.e., self-attention heads), to reduce final model size.

DiffLoRA technique is a multi-step operation that utilizes LoRA and two images: original image A and manipulated image B. Initially, we fine-tune the original large model with image A using LoRA until it over-fits, resulting in Lora A. We combine this over-fitting Lora A with the original model to create model A. Subsequently, we repeat the same process using model A (not the original model) and image B, generating the over-fitting LoRA B and model B. Finally, we use the discrepancy between model B and the original model to establish DiffLoRA. This technique can effectively capture and learn the critical differences between the two inputs as the desired attribute range of continuous values.

Combining these two techniques, *Semantic Injection* can efficiently encode all types of attributes into the large model, the attributes in discrete values through Fast Lora and continuous values through DiffLora, respectively. The refinement over a large model can acquire understanding and effective synthesis of a given artistic style according to its semantic description guideline. Incorporating the various fine-tuned LoRAs with the original model as a customized model, *Semantic Injection* can lead to *Artist Model* in any given artistic style.

²Fast LoRA: <https://github.com/cloneofsimon/lora>

³DiffLoRA: <https://pixai.art/model/1630413070130926806>

3 IMPLEMENTATION

To systematically describe Kandinsky’s Bauhaus style as a semantic descriptive guideline, we collaborate with researchers specializing in Kandinsky’s art [Zhang and Yu 2013][Zhang and Yu 2016] and examine two books of Kandinsky’s theory [Kandinsky 1977] [Kandinsky 1979]. We summarize the literature and derive a list in Fig.3, consisting of 7 attributes and 22 values. Furthermore, we select and categorize 65 representative artworks through expert evaluation and finally label the selected paintings using our semantic descriptive guideline. The labels start with a prefix of “Kandinsky,” followed by attribute values. We selectively retain the attributes that have discernible effects and omit those that are non-obvious or controversial, which ensures a well-balanced dataset across various attributes and encompasses distinct and explicit Kandinsky features.

Attribute		Value					
Color	Brightness	Light				Dark	
	Hue	White		Grey		Black	
		Orange	Yellow	Red	Green	Blue	Violet
Form	Point					Point	
	Line	Curve Line		Angular Line		Straight Line	
	Plane	Triangle		Square		Circle	
Composition	Structure	Acentric Structure				Centric Structure	
	Parallel	External Parallel				Inner Parallel	

Figure 3: Semantic descriptive guideline: Kandinsky style

In our experiment of *Semantic Injection*, we employ fast LoRA to inject hue and form-related attributes and evaluate the performance with Kandinsky expert by using attribute value as single-keyword prompts to generate images and subjectively validate the results. For continuous values such as color brightness and composition-related attributes, we apply DiffLoRA, deriving four additional LoRA models. Together with the base model, it results in the *Artist Model* for Kandinsky Bauhaus style generation that users can work with as a deterministically controllable experience. Fig.1 shows prompting the *Artist Model* following the semantic descriptive guideline.

4 DISCUSSION AND LIMITATION

The current experiment is mostly artist and expert-oriented and lacks quantitative evaluation. A comprehensive study involving users’ insights is needed for further improvements. Future research can automate the establishment of the semantic descriptive guideline, as it now heavily relies on expert knowledge and manual examination of literature, which limits its efficiency and scalability.

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